

Challenge 1: Beat your benchmark

MATH840 | Time Series | Week 4 — graded assignment 3 of 7

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! What changed this week

You asked to do the graded work at home, and that is how it works from now on.

- **The practice session is no longer the lab.** We spend it on worked examples and discussion — the moves you are about to need, on a shared dataset, with everything asked out loud.
- **The graded work is done at home, due 23:59 on the day of the practice session.** No in-class submission, no baseline deadline.
- **Attendance is not marked.** The session is worth coming to because the examples are the ones you will reuse an hour later, not because a register is taken.

💡 Open the notebook

 [Open in Colab](#)

[Open In Colab](#)

Click **Copy to Drive** before you start working, otherwise nothing you write is saved.

🎯 The task

This is the first of five **challenges**. From today you have your own time series, and one number to beat on it.

Forecast the next h observations of your series and beat your benchmark — with the toolbox from Week 3 and nothing more. No ETS, no ARIMA: those are the next challenges, and this one is the measurement that makes their improvement visible.

! This one counts

8 points. **Deadline: 23:59 today**, on Moodle. The accuracy score is published the next morning; rubric marks and comments come back before the next practice session.

📁 Your series

Your series is issued by a hash of the **same identifier you have used since Week 1**. The template does it for you: it turns your identifier into a code such as `TS-4DD7A` and downloads that series.

i What “anonymised” means here

The series is real. What has been removed is its identity: no name, no units, values rescaled, the calendar shifted by whole years, the history trimmed. Two students never get the same window of the same series, so a forecast is worth exactly what your model is worth.

You are not expected to work out what it measures, and you will not be marked for trying. Read the data, not the domain.

This series is yours for the rest of the course. Challenges 2–5 (ETS, ARIMA, seasonal ARIMA, advanced methods) run on the same series, so you will watch your own forecast improve — or fail to.

What you are marked on

Section	Criteria	Points
1	EDA and visualisation	1
2	Forecast and accuracy	3
3	Justification block	3
4	Code quality and reproducibility	1
Total		8

The notebook has one section per row. **Keep those headings.**

Section 1 — EDA and visualisation (1 point)

One point, and it is not a repeat of Lab 2: this is a **new series**, and the plots are here to support the choice of method.

- A time plot, and whichever of the seasonal / subseries / ACF plots your argument actually uses.
- Every plot needs a sentence. A plot with no reading attached earns nothing.
- Show the validation split on a plot: history, validation window, and your forecast over it.

Section 2 — Forecast and accuracy (3 points)

Your benchmark

The benchmark is the **best of the four simple methods** from Week 1 — `mean`, `naive`, `snaive`, `drift` — chosen on a validation window: the last h observations of the history you were given.

Nothing about it is hidden. The template computes it, and the scorer computes it the same way, from the same public data:

- 1. fit the four methods to everything except the last h observations;
- 2. score them on those h observations by RMSE, and keep the winner;
- 3. refit the winner on the **whole** history you were given — that is your benchmark.

How the points are awarded

Your forecast and the benchmark’s are both scored by **MASE** on the h observations that come after your history, which you have not seen.

$$\text{skill} = \frac{\text{MASE}_{\text{you}}}{\text{MASE}_{\text{benchmark}}}$$

Result	Points
$\text{skill} < 1.00$ — you beat the benchmark	3
$\text{skill} \leq 1.10$ — within 10% of it, or equal to it	2
worse than that	1, and Section 3 must say why

i Your series is known to be beatable

Every series in the pool was screened: it is only issued if a reference model lands between **0.70 and 0.90** of its benchmark. Below that the series is trivial, above it effectively unbeatable. If you do not beat your benchmark, that is a fact about your model, not your luck.

The benchmark is also checked for being a fair target: series where one of the other three simple methods would have beaten the chosen one by more than 15% on the hidden holdout were thrown out, so you are not competing against an unlucky pick.

The screening reference was an automatic ETS or ARIMA — models you meet in the next two challenges. With the toolbox alone the odds are tighter, and they have been measured on this pool: a careful attempt — several candidates, chosen on the validation window — **beat the benchmark on about a third of the series, came within 10% on nearly half, and lost on the rest.**

So two accuracy points is the typical outcome this week, not a failure, and the way to three is a better argument rather than a luckier guess. In Challenge 2 the same series meets ETS, and the odds change: that is the improvement this week’s number exists to measure.

What the toolbox gives you

Everything in this list is Week 3 material, and each of them can beat a simple method:

- a **transformation** — Box-Cox, if the seasonal swing grows with the level;
- **forecasting with decomposition**: STL the series, forecast the seasonally adjusted part with **naive** or **drift**, carry the seasonal component forward with **snaive**, add them back;
- a **combination**: the average of two or three forecasts is often better than any of them;
- **residual diagnostics** to catch a model that is leaving structure behind.

Section 3 — Justification block (3 points)

Three points, and the only part of the work a human reads closely. Four things, roughly a paragraph each:

Item	Points	What earns them
3.1 What the data shows	0.75	Frequency, seasonal period, trend, level shifts, outliers — with the plot that shows each claim
3.2 Why this method	0.75	The method you submitted and why it fits what you just described , including what you tried and dropped
3.3 What the residuals say	0.75	ACF and a Ljung-Box test on your final model's residuals, read — what structure is left, and whether it matters at this horizon
3.4 What you expect	0.75	Your own MASE on the validation window against the benchmark's, and the number you expect on the hidden holdout

Important

Marked on the reasoning, not on the outcome. A well-argued forecast that loses to the benchmark scores full marks in Section 3 — and if it loses, say what you think the holdout contained that your validation window did not.

Section 4 — Code quality and reproducibility (1 point)

- The notebook runs top to bottom on a fresh kernel without errors.
- Your submission file is written **by the notebook**, not edited by hand.
- No leakage: nothing after the end of your history exists, so nothing can be fitted to it. If you tune on the validation window, say so in 3.2 — that is legitimate, and it is also why your validation MASE is optimistic.

Submission

Upload three files to the Week 4 activity on Moodle **before 23:59 today**:

- `SURNAME_ch1_forecast.csv` — written by the last cell of the template. Exactly h rows and these columns:

Column	What
<code>student_id</code>	your identifier
<code>code</code>	your series code, e.g. <code>TS-4DD7A</code>
<code>ds</code>	the dates you are forecasting, in order
<code>yhat</code>	your point forecast
<code>yhat_lo_80</code> , <code>yhat_hi_80</code>	your 80% interval — not marked this week, but the scorer records its coverage

- `SURNAME_ch1.pdf` — the report, with the justification block;
- `SURNAME_ch1.ipynb` — the source.

 **The scorer is strict about the format, and only about the format**

Wrong dates, wrong number of rows, a missing column or a `NaN` in `yhat` cannot be scored, and Section 1 is then 0. The template writes the file correctly; if you rebuild it by hand, run the check cell it provides.

How long this should take

Ninety minutes if you follow the practice session's example, half of that if you reuse its code. The work is deliberately not bigger than a session — it moved home so that nobody has to produce a forecast and an argument in a room with a clock, not so that it could grow.

For next week

Week 5 is the mid-term quiz — no practice session and no challenge. It covers Weeks 1–4: forecasting basics and graphics, decomposition, the toolbox, exponential smoothing.

Challenge 2 is in Week 6, on this same series, with ETS — which is the Week 4 lecture, already published. Read [FPP3 ch. 8](#) before then and your second challenge is half done.

